Workshop 2

Design Artifacts and System Modeling

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1. System Domain

- **System Context:**

What is VetCare?

VetCare is a centralized web platform for veterinary clinics. It brings client records, clinical records, appointments, hospitalization and pharmacy inventory into a single system, with access restricted by role. It is organized into four modules:

- Authentication & Access Control: accounts, sessions and permissions per role.

- Patient & Client Management: owner and pet records, species-specific clinical records, vaccinations and prescriptions.

- Space & Appointment Management: online booking, room and veterinarian scheduling, hospitalization map and automated reminders.

- Pharmacy & Inventory Management: stock by batch, automatic deduction, low-stock and expiration alerts, and suppliers.

Hospitalization belongs to the Space & Appointment Management module. In the following sections it is presented as a separate group for clarity, so the classes and business rules are organized into five groups.

Who uses it?

VetCare is used by two groups:

-Clinic staff (Veterinarians, Receptionists, Veterinary Assistants and Administrators) use it for daily operations.

-Pet owners use a health portal to request and manage appointments, check a simplified medical summary of their pets and receive reminders.

The system is accessed through a web browser on desktop, tablet and mobile devices, and its interface and notifications are in Spanish by default.

What problem does it solve?

Veterinary clinics often manage client data, clinical records, appointments, hospitalized patients and medication stock in separate tools, such as paper files, spreadsheets, phone calls and messaging apps. This fragmentation makes it hard to retrieve the history of a patient, avoid scheduling conflicts, control medication stock and keep owners informed.

What is the scope?

The system includes the four modules described above and the requirements defined in Workshop 1. It interacts with an external messaging service to deliver confirmations and reminders to owners through their preferred channel. The following aspects are outside the scope of this project: billing and payments, accounting, and the management of multiple clinics or branches.

- **Actor Characterization:**

The actors of VetCare are the users and external systems that interact with it. Five are user actors (four clinic staff roles and the pet owner) and one is an external service. The permissions of each user actor are enforced through role-based access control (RBAC).

Actor	Description	Main activities	Restrictions
Pet Owner	Client of the clinic who owns one or more pets. Uses the health portal with an account created by registration or by activating an invitation from the clinic.	-Requests, cancels and reschedules appointments online. -Views the available time slots. -Views a simplified medical summary of each pet. -Receives appointment confirmations, appointment reminders and vaccination reminders.	-Cannot view internal veterinary notes. -Cannot access any staff function.
Receptionist	Staff member in charge of client registration and appointment management.	-Registers owners, sends portal invitations and links pets to owners. -Searches owners and patients by owner name, national ID or pet name. -Receives notifications of requests, cancellations and reschedules. -Confirms, reschedules and cancels appointments. -Assigns consultation rooms without double-booking rooms or veterinarians.	-Cannot record clinical data, issue prescriptions or manage inventory.

Veterinarian	Professional who attends patients and is responsible for their clinical records and hospitalization.	-Opens species specific consultation files and records vital signs, diagnoses and treatments. -Issues prescriptions linked to inventory items. -Views the complete clinical record of a patient. -Records vaccine applications and supplies consumed. -Admits and discharges hospitalized patients, changes their status and records round observations. -Views the log of status changes.	-Cannot manage staff accounts, schedules, purchases or suppliers.
Veterinary Assistant	Staff member who supports the veterinarian in routine care and in the monitoring of patients.	-Views the complete clinical record of a patient. -Records vaccine applications and views their next due dates. -Records hospitalization round observations and supplies consumed during patient care.	-Cannot issue prescriptions, admit or discharge patients, or change the status of hospitalized patients.
Administrator	Staff member responsible for the configuration and the resources of the clinic.	-Creates, edits and deactivates staff accounts and assigns roles. -Defines the weekly working hours and time off of veterinarians. -Registers purchase receipts and manages inventory items and suppliers. -Receives low-stock and expiration alerts and views the contact details of the supplier.	-Does not take part in clinical activities such as consultations or prescriptions.

Messaging Service (external)	External e-mail service used by the system to deliver automated messages. Other channels such as SMS and WhatsApp can be integrated later.	-Delivers appointment confirmations, appointment reminders and vaccination reminders to owners through their preferred channel.	-Does not access clinical data beyond the content of each message.

- **Business rules:**

The business rules are the policies and constraints of the clinic that the system must enforce. The rules are grouped by system module.

ID	Module	Business rule
BR_01	Access Control	Every user has one or more roles (Veterinarian, Receptionist, Administrator, Veterinary Assistant or Owner). The permissions of a user are the union of the permissions of all assigned roles, and the user can only perform the actions allowed by them.
BR_02	Access Control	Only an Administrator can create, edit and deactivate staff accounts and assign their roles.
BR_03	Access Control	A deactivated account cannot log in.
BR_04	Access Control	A change in the roles of a user applies the new permissions from the next login of the user.
BR_05	Access Control	An account is locked for 15 minutes after 5 consecutive failed login attempts.
BR_06	Access Control	A session ends after 30 minutes of inactivity, and a new login is required.
BR_07	Access Control	A password reset link is time-limited and is sent only to the registered email of the user.
BR_08	Patient & Client	Every pet belongs to an owner, and one owner can have several pets under a single customer record.
BR_09	Patient & Client	Each owner has a preferred communication channel, which is used for all automated messages.
BR_10	Patient & Client	A consultation file uses the medical fields specific to the species of the patient.
BR_11	Patient & Client	Clinical entries cannot be deleted. A correction is stored as a new version that preserves the previous one.
BR_12	Patient & Client	Every view or modification of a clinical record is logged with the user, the action and the timestamp.

BR_13	Patient & Client	A Pet Owner can only see a simplified medical summary of the pet and cannot see internal veterinary notes.
BR_14	Patient & Client	A vaccine application records the vaccine, the lot number and the application date. The next due date is calculated automatically from the application date.
BR_15	Patient & Client	A prescription includes the drug, dose, frequency and duration, and is linked to an inventory item.
BR_16	Appointments	An appointment requested online by an owner is registered as pending until a Receptionist confirms it.
BR_17	Appointments	A consultation room and a veterinarian cannot be double-booked in the same time slot.
BR_18	Appointments	Only time slots within the working hours of a veterinarian, excluding time off, are offered for appointments.
BR_19	Appointments	An owner can cancel or reschedule only appointments in pending or confirmed status, and can reschedule only to an available time slot.
BR_20	Appointments	When an appointment is confirmed, rescheduled or cancelled, the available time slots are updated.
BR_21	Appointments	The reception team is notified when an owner requests, cancels or reschedules an appointment.
BR_22	Appointments	The owner receives a confirmation when an appointment is confirmed, a reminder when its date approaches, and a vaccination reminder when the next due date of the pet approaches.
BR_23	Hospitalization	A patient can only be admitted to a free cage. The cage is marked as occupied until the patient is discharged.
BR_24	Hospitalization	A hospitalized patient has one status at a time (Critical, Stable or Under Observation), and the initial status is set at admission.
BR_25	Hospitalization	Every status change and every round observation is recorded with its author and timestamp.
BR_26	Hospitalization	A discharge records the discharge date and a summary, and releases the cage.
BR_27	Pharmacy & Inventory	Stock is managed by batch, with a lot number and an expiration date.
BR_28	Pharmacy & Inventory	Stock is added to the inventory only by registering a purchase receipt with items, quantities, lot numbers, expiration dates and supplier.
BR_29	Pharmacy & Inventory	Prescribed drugs, applied vaccines and consumed supplies are deducted automatically from the inventory, and every deduction is an atomic transaction.

BR_30	Pharmacy & Inventory	Every inventory item has a minimum threshold and a linked supplier.
BR_31	Pharmacy & Inventory	An alert is generated when an item drops below its minimum threshold or reaches its expiration period.

2. CRC Cards

- **Identified Classes:**

The classes below were identified from the functional requirements and user stories of Workshop 1. Each class represents a concept of the clinic domain that stores data and performs behavior. They are grouped by system modules: access control, patient and client management, appointments, hospitalization, and pharmacy and inventory.

Module	Class	Description
Access Control	User	Account with email, password and status. Logs in and out of the system
Access Control	Role	Veterinarian, Receptionist, Administrator, Veterinary Assistant or Owner, with the permissions of each one
Patient & Client Management	Owner	Personal details and preferred communication channel
Patient & Client Management	Pet	Patient with species, allergies and owner
Patient & Client Management	ConsultationRecord	Consultation with vital signs, diagnosis and treatment. Corrections are stored as a new version
Patient & Client Management	Prescription	Drug, dose, frequency and duration
Patient & Client Management	Vaccination	Applied vaccine, lot, date and next due date
Patient & Client Management	AuditLog	User, action and timestamp of every view or modification of a clinical record
Appointments	Appointment	Visit with status (pending, confirmed, cancelled), pet, service type and time slot
Appointments	VeterinarianSchedule	Weekly working hours and time off of a veterinarian
Appointments	Room	Consultation room assignable to an appointment
Appointments	Notification	Confirmation and reminder messages sent through the preferred channel
Hospitalization	Hospitalization	Admission with status, discharge date and discharge summary
Hospitalization	Cage	Cage or bed shown on the occupation map

Hospitalization	RoundObservation	Observation of a hospitalization round, with author and timestamp
Hospitalization	StatusChange	Record of each status change (Critical, Stable, Under Observation)
Pharmacy & Inventory	InventoryItem	Drug, vaccine or supply with its minimum threshold
Pharmacy & Inventory	Batch	Lot with quantity and expiration date
Pharmacy & Inventory	Supplier	Supplier contact details
Pharmacy & Inventory	PurchaseReceipt	Purchase that adds batches to the inventory
Pharmacy & Inventory	StockAlert	Alert for low stock or approaching expiration

- **CRC Cards:**

Class: User	
Responsibilities	Collaborators
Store account credentials and account status	Role
Authenticate credentials on login and establish the active session; deny access on invalid credentials	
Hash sensitive credentials and validate JWT tokens on each request	
Be created at owner registration or by an Administrator who assigns roles	Role, Owner
Identify the acting account when clinical records are viewed or modified	AuditLog

Class: Role	
Responsibilities	Collaborators
Define the permission set of each role (Veterinarian, Receptionist, Administrator, Veterinary Assistant, Owner)	
Answer whether a given user may perform a given action	User

Restrict Owners to records of pets linked to their own client record	Owner, Pet

Class: Owner	
Responsibilities	Collaborators
Store the owner's personal details and preferred communication channel	
Receive confirmations and reminders through the preferred channel	Notification
Be linked to the user account used in the health portal	User
Guarantee that its access is scoped to only its own pets	Pet, Role

Class: Pet	
Responsibilities	Collaborators
Store the patient profile (species, allergies, owner link)	Owner
Provide the simplified data for the owner's health portal	Vaccination, ConsultationRecord, Prescription
Provide the complete medical background: vaccination history, allergies, previous consultations	Vaccination, ConsultationRecord

Class: ConsultationRecord	
Responsibilities	Collaborators
Store the vital signs, diagnosis and treatment of a consultation, with the fields specific to the species	Pet
Store corrections as new versions, preserving the clinical history	
Display the complete clinical record to Veterinarians and Assistants	Pet, Prescription, Vaccination
Render the simplified, jargon-free medical summary for the owner portal, excluding	Pet, Prescription, Vaccination

internal notes	
Leave a trace of every view or modification	AuditLog, User

Class: Vaccination	
Responsibilities	Collaborators
Record the applied vaccine, lot number, date and next due date	Pet, User
Expose vaccination status and next due date for the owner portal	Pet
Provide the complete vaccination history for authorized searches	Pet
Trace the applied lot back to the pharmacy stock for traceability	Batch
Indicate when a vaccination reminder is due	Notification

Class: Prescription	
Responsibilities	Collaborators
Record the drug, dose, frequency and duration prescribed by the Veterinarian	
Trigger the automatic deduction of the prescribed quantity from stock	Batch, InventoryItem
Appear among the active treatments shown in the owner portal	

Class: AuditLog	
Responsibilities	Collaborators
Record who did what and when, for every access to or modification of a clinical record	User, ConsultationRecord
Serve as the evidence trail for corrections stored as versions	ConsultationRecord

Class: Appointment

Responsibilities	Collaborators
Log the owner's online booking request in "Pending" status with pet, service type and time slot	Pet, Owner
Update its state when the Receptionist confirms, reschedules or cancels	
Check doctor and room availability before confirming; reject the conflict and suggest the next available slots	VeterinarianSchedule, Room
Trigger the confirmation/reminder notification when confirmed or when the date approaches	Notification

Class: VeterinarianSchedule	
Responsibilities	Collaborators
Store each veterinarian's weekly working hours and approved time off	User
Answer availability queries for a given time slot	Appointment
Provide the next available time slots if the requested one is booked	Appointment

Class: Room	
Responsibilities	Collaborators
Check room availability per time slot	Appointment
Reject conflicting assignments (double-booking)	Appointment
Provide room availability. If a slot is already booked, provide the next available slot.	Appointment

Class: Notification	
Responsibilities	Collaborators
Deliver the automated confirmation when an appointment is confirmed	Appointment, Owner

Deliver the reminder when the appointment date approaches	Appointment, Owner
Send all messages through the owner's preferred communication method	Owner
Deliver the vaccination reminder when the next due date approaches	Vaccination, Owner
Notify the reception team when an owner requests, cancels or reschedules an appointment	Appointment, User

Class: Hospitalization	
Responsibilities	Collaborators
Manage the admission status: admission, evolution, discharge	Pet
Manage the patient's current status (Critical, Stable, Under Observation)	StatusChange
Collect round observations into the patient's evolution log	RoundObservation
Free the occupied cage/bed at discharge	Cage, StatusChange

Class: Cage	
Responsibilities	Collaborators
Report its occupation status indicator (Free, Critical, Stable, Under Observation)	Hospitalization
Reflect status changes on the patient's indicator	StatusChange
Accept/release a hospitalized patient	Hospitalization

Class: RoundObservation	
Responsibilities	Collaborators
Record the staff member's ID, timestamp and observations of the round	User, Hospitalization
Serve timestamped notes into the patient log	Hospitalization

Class: StatusChange	
Responsibilities	Collaborators
Store each status transition with its timestamp to maintain an accurate evolution log	User, Hospitalization
Update the patient status indicator	Hospitalization, Cage

Class: InventoryItem	
Responsibilities	Collaborators
Store each drug/vaccine/supply and its minimum stock threshold	
Display available quantities categorized by batch/lot and expiration date	Batch
Raise an alert when quantity drops below the threshold or nears expiration.	Batch, StockAlert
Present the supplier contact details linked to the item for reordering	Supplier

Class: Batch	
Responsibilities	Collaborators
Track quantity and expiration date per lot	InventoryItem
Deduct consumed or prescribed quantities automatically	Prescription, Vaccination
Notify corresponding alerts for registered expiration periods	StockAlert, InventoryItem
Grow when a purchase adds the lot to stock	PurchaseReceipt

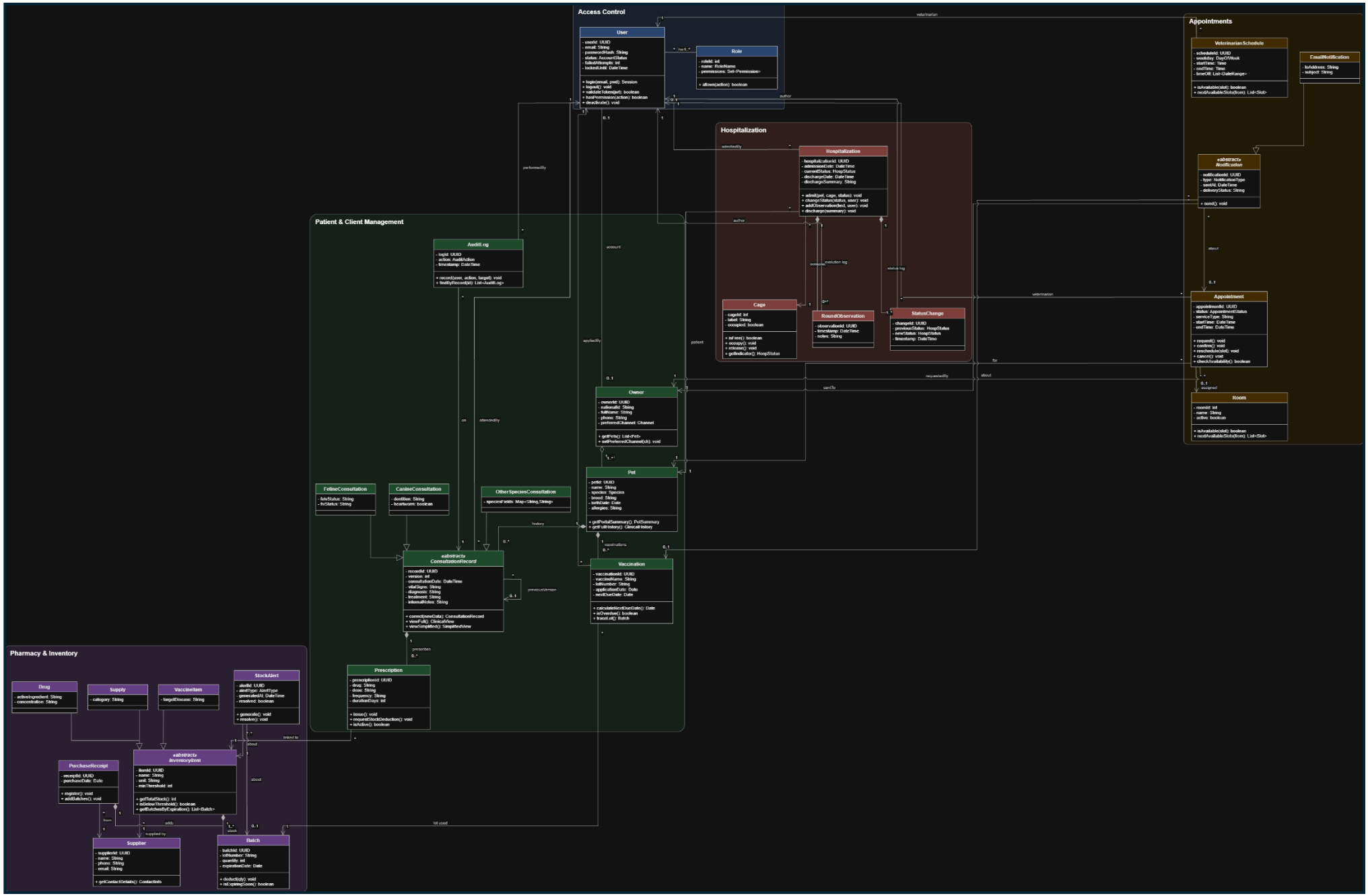
Class: Supplier	
Responsibilities	Collaborators
Store supplier contact details for each item	InventoryItem
Present its contact details when an	StockAlert

Administrator reviews a low-stock entry	

Class: PurchaseReceipt	
Responsibilities	Collaborators
Register an incoming purchase from a supplier	Supplier
Add the purchased batches (lots, quantities, expiration dates) to the inventory	Batch, InventoryItem

Class: StockAlert	
Responsibilities	Collaborators
Flag medications/vaccines that drop below its threshold or nears expiration	Batch, InventoryItem
Highlight low-stock items with an alert badge on the dashboard	InventoryItem
Provide direct access to the supplier contact details of alerted items	Supplier

3. Class Diagram



Link:
https://drive.google.com/file/d/1AsDpG0USutgJkR9Mi_zClAmGWZDLVW6q/view?usp=drive_link

Reading the Diagram

Each box is a class with its private attributes (marked with "-") and its public methods (marked with "+"). A solid line with a hollow triangle is inheritance, a filled diamond is composition, a hollow diamond is aggregation, and an open arrowhead is a directed association. The numbers at each end of a line are multiplicities. The background color of each group identifies the system module.

Overview

The model contains 28 classes: the 21 identified in the CRC cards and seven specializations added to make the business rules explicit. These are CanineConsultation, FelineConsultation and OtherSpeciesConsultation (species-specific consultation files), Drug, VaccineItem and Supply (types of inventory item), and EmailNotification (the first delivery channel). The classes are grouped into the same five modules used in the previous sections.

Main Design Decisions

- Separate account, client record and role. User, Owner and Role are different concepts. This allows owners without a portal account and users with several roles (BR_01), and keeps permission checks independent of the type of person.

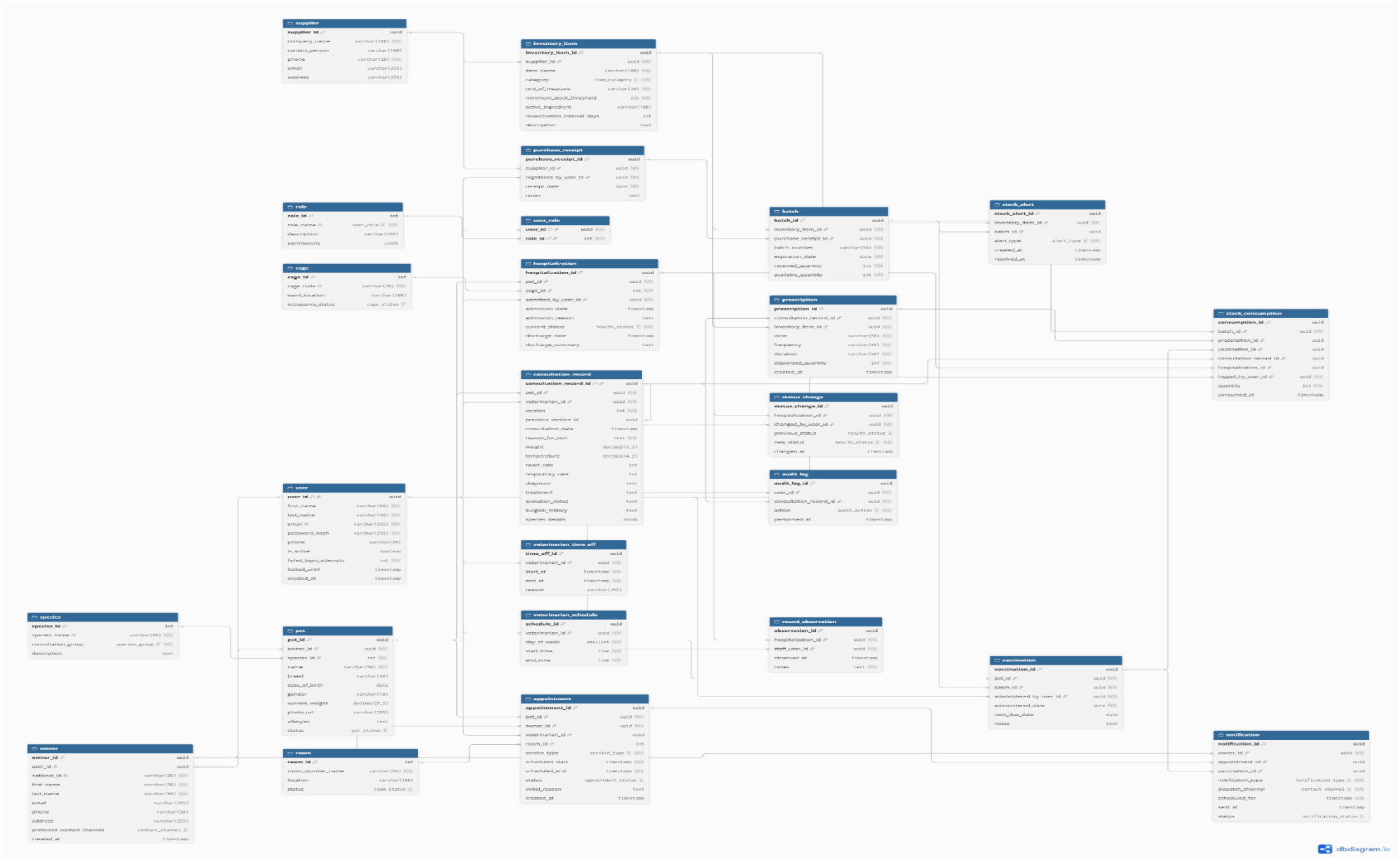
- Inheritance only where fields or behavior change by type. ConsultationRecord (by species, BR_10), InventoryItem (by kind) and Notification (by channel) have a common core and type-specific parts. A new species, supply type or channel such as SMS can be added as a subclass without changing the rest of the model.

- Composition for histories and logs. Consultations, vaccinations, status changes, round observations and batches cannot exist without their parent. Owner and Pet use aggregation because a pet is an independent patient.

- Traceability by design. Versioned consultation records (BR_11), the AuditLog (BR_12), the author of every observation and status change (BR_25) and the link from Vaccination to Batch make it possible to reconstruct who did what, when and with which lot.

- Stock consumed at lot level. Prescriptions and vaccinations deduct from a specific Batch, which allows the atomic transaction required by BR_29.

4. Relational Database Model



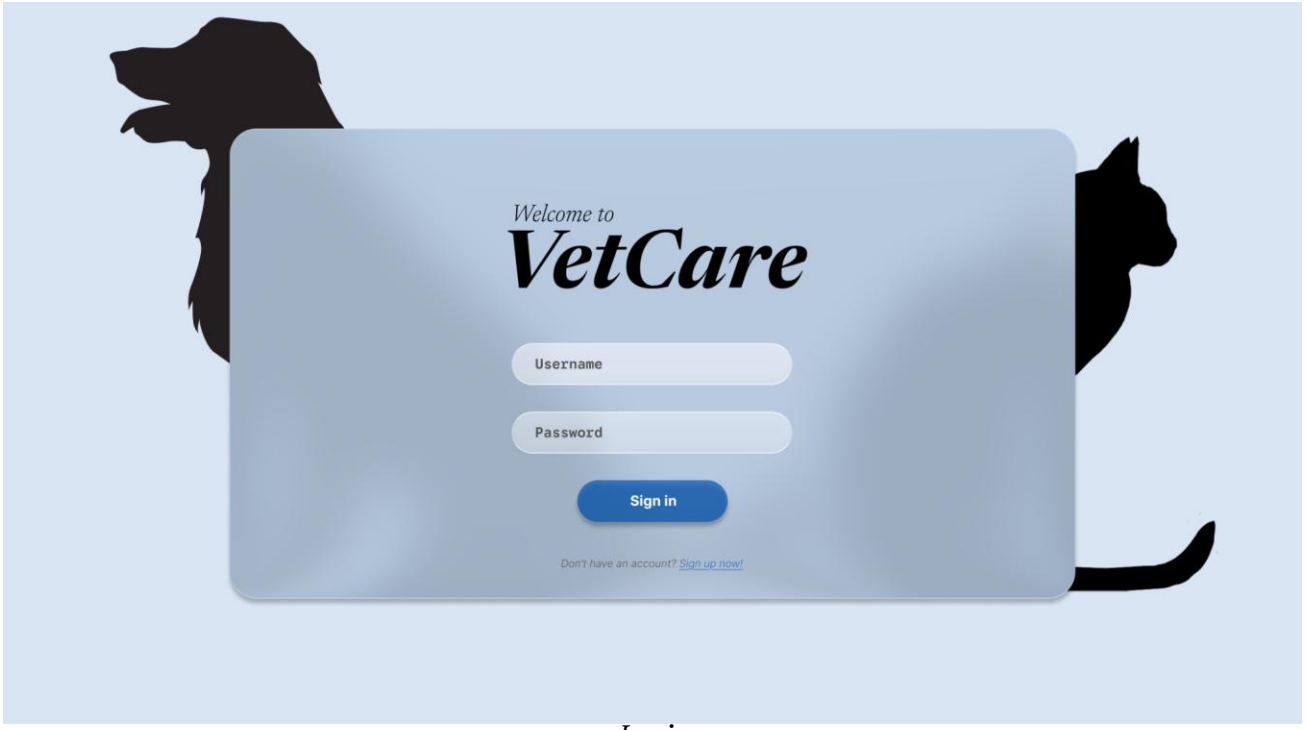
Link: https://dbdiagram.io/d/VetCare-69f1d0e2ddb9320fdc88d8a6

The relational model contains 25 tables and follows the class model: each class maps to a table with the same name in snake_case. The specializations (the species-specific consultations, the inventory item types and the email notification) are stored in the table of their parent class.

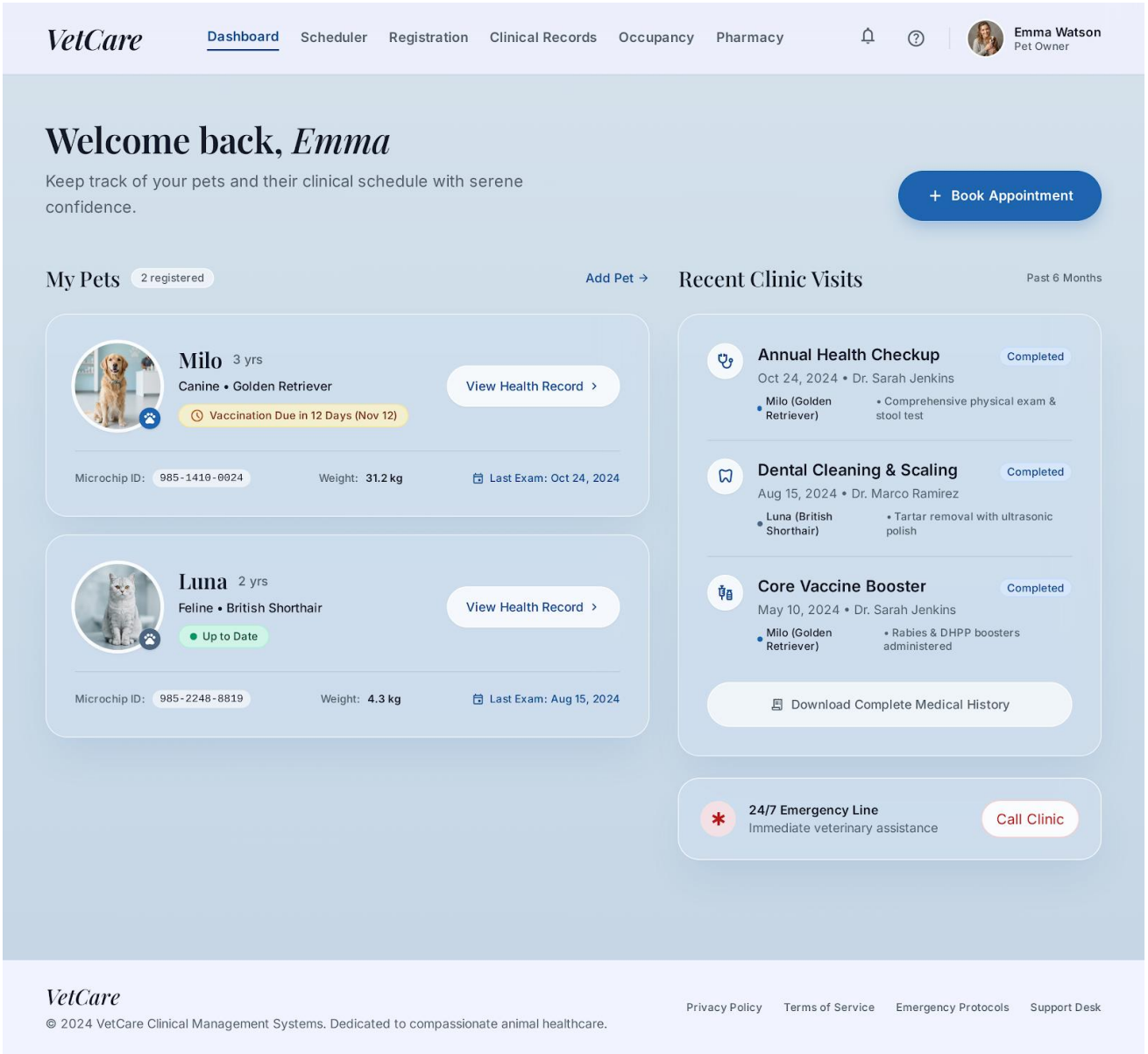
The many-to-many relationship between User and Role uses the join table user_role. A correction of a clinical entry creates a new consultation_record row linked to the previous one, so no clinical entry is deleted. Stock is deducted at lot level through stock_consumption, which links each batch to the prescription, vaccination, consultation or hospitalization that consumed it.

5. Mockups

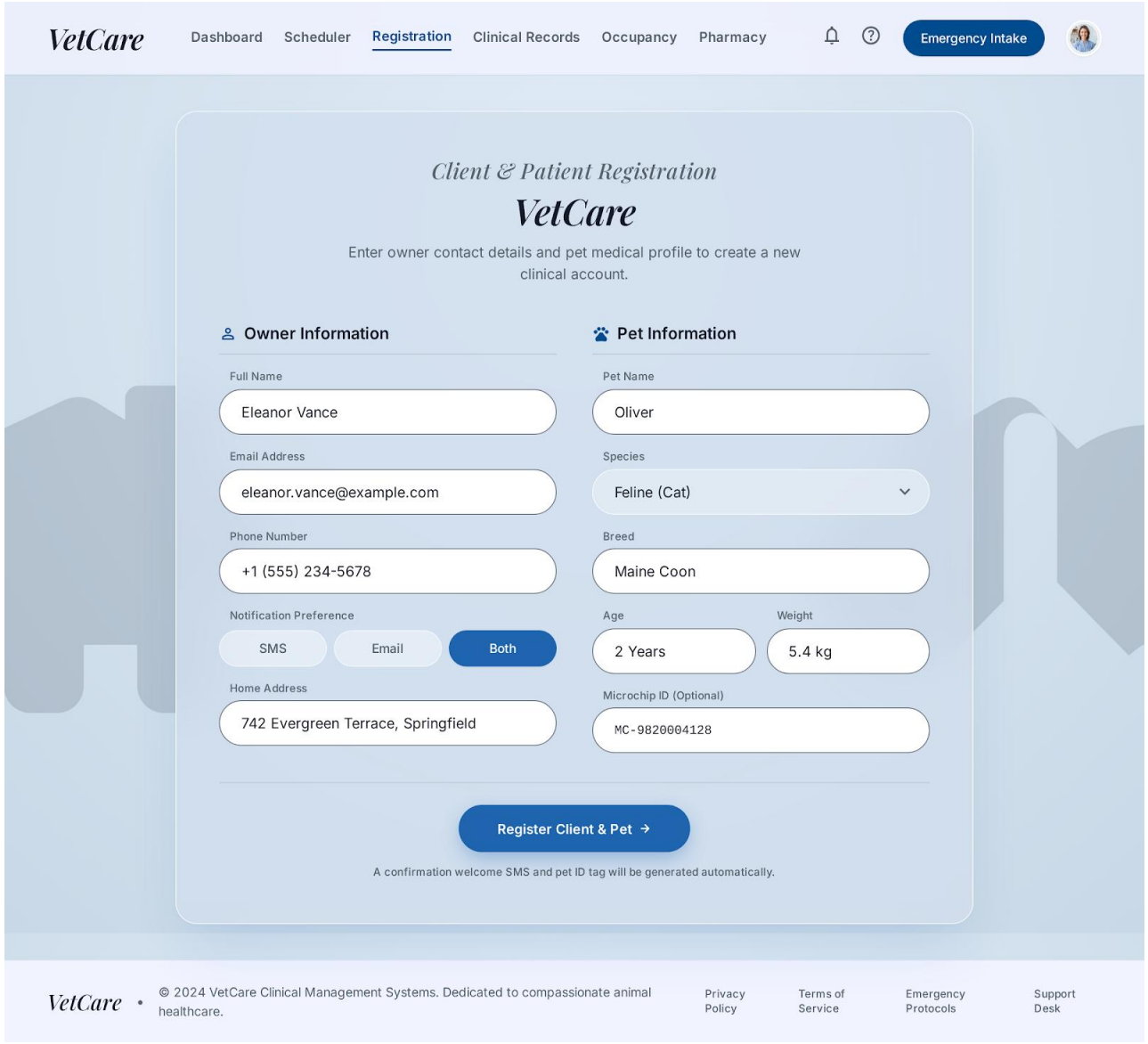
- **Login and Registration Screen:** Provides general access to the system based on assigned roles (RBAC) and includes a link for pet owners to create an account.
- **Pet Owner Dashboard (Simplified Health Portal):** A simplified view for clients displaying pet profiles, vaccination status with upcoming due dates, recent clinic visits, and a direct button to request appointments.
- **Owner and Pet Registration Form:** A reception form to register new clients with their contact details and link one or multiple pets to a single master profile.
- **Visual Hospitalization Map and Rounds Log:** An interactive visual grid representing cages/beds with color-coded status indicators (Free, Critical, Stable, Under Observation). Selecting a cage opens a side drawer to log round checks and discharge the patient.
- **Clinical Record and Medical Consultation (Veterinarian View):** A comprehensive clinical file for veterinarians featuring a species selector (dogs, cats, exotics) that adapts the clinical fields for vital signs, diagnoses and prescribed treatment.
- **Appointment and Room Management Module (Receptionist Dashboard):** A receptionist interface featuring a daily calendar by veterinarian (pending, confirmed, canceled), a list of pending requests, and a dialog to assign a consultation room and veterinarian designed to prevent scheduling overlaps.
- **Pharmacy and Inventory Dashboard:** An administrative view displaying stock levels categorized by batch number and expiration date, automated low-stock and expiration alerts, and the contact details of the supplier of the selected item.


Welcome to
VetCare
Username
Password
Sign in
Don't have an account? Sign up now!

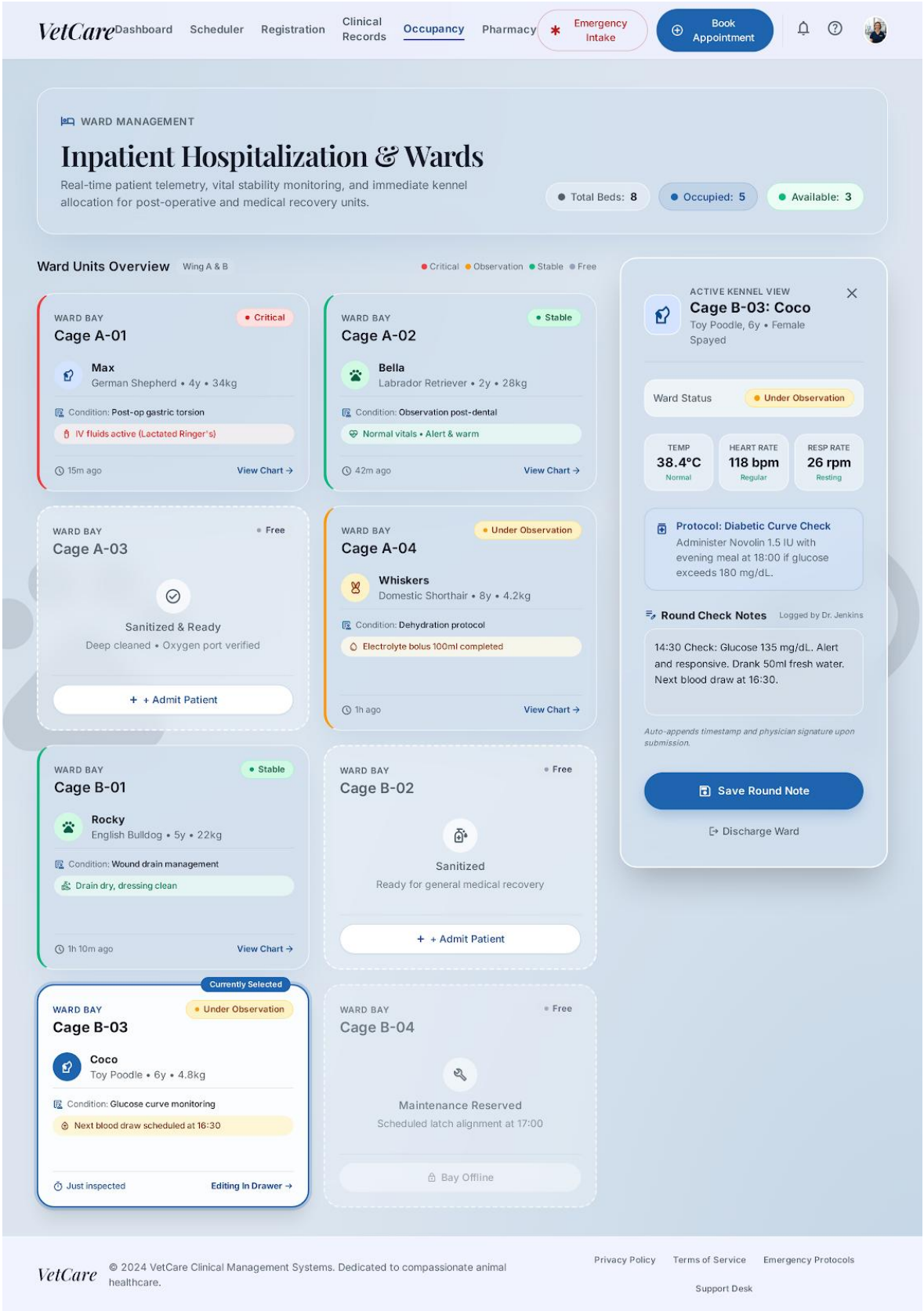
Login



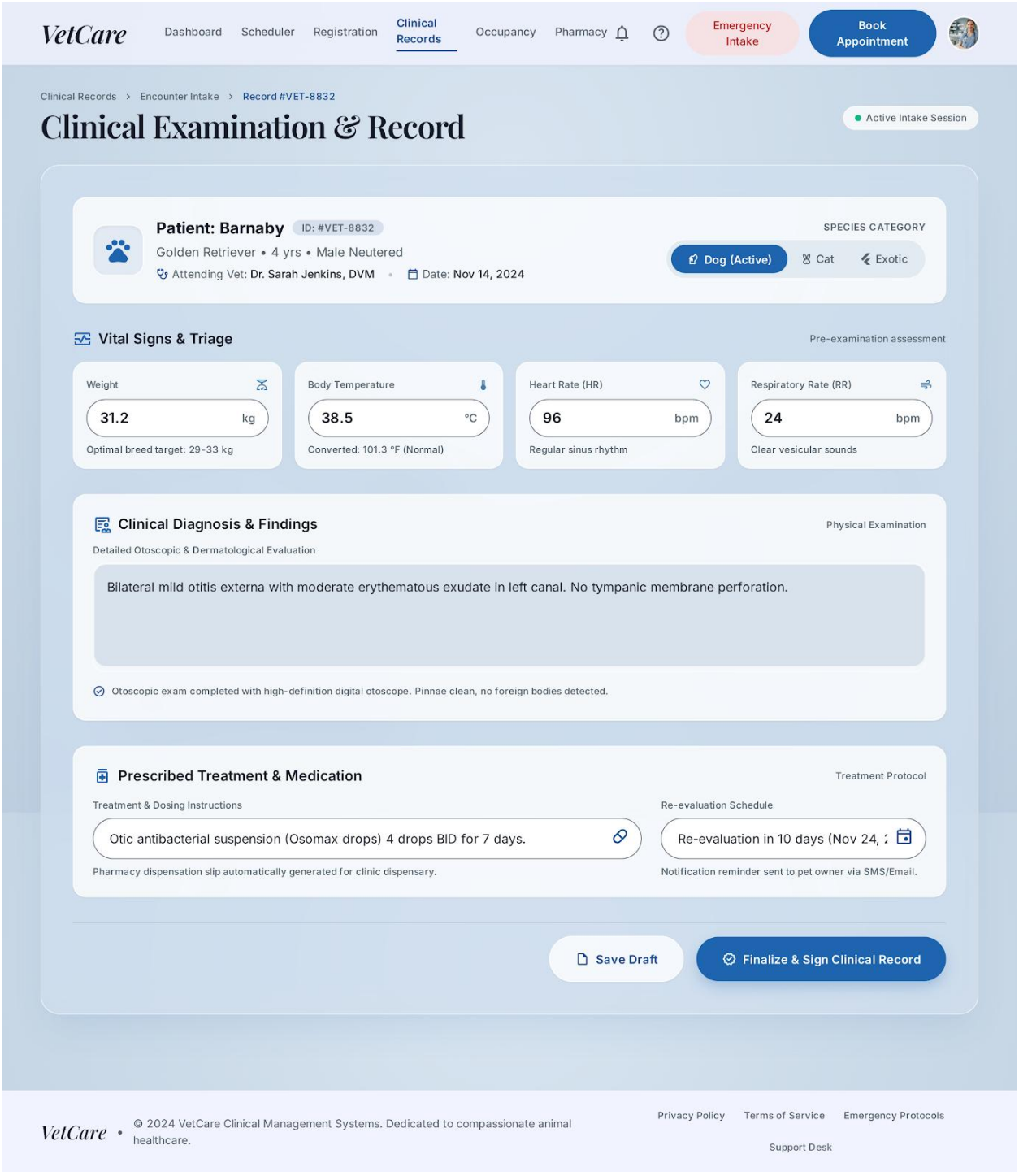
Pet Owner Dashboard



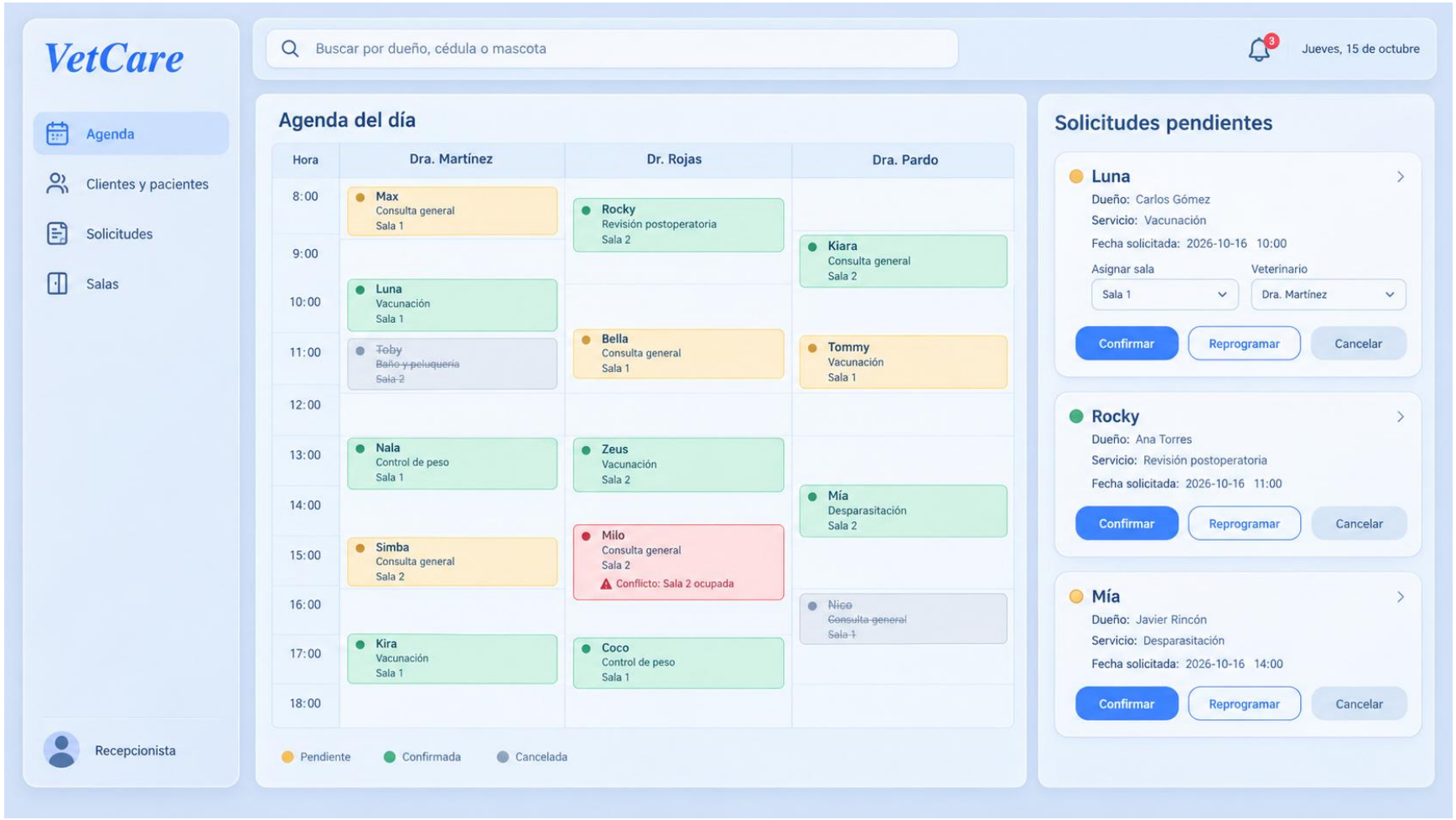
Owner and Pet Registration Form



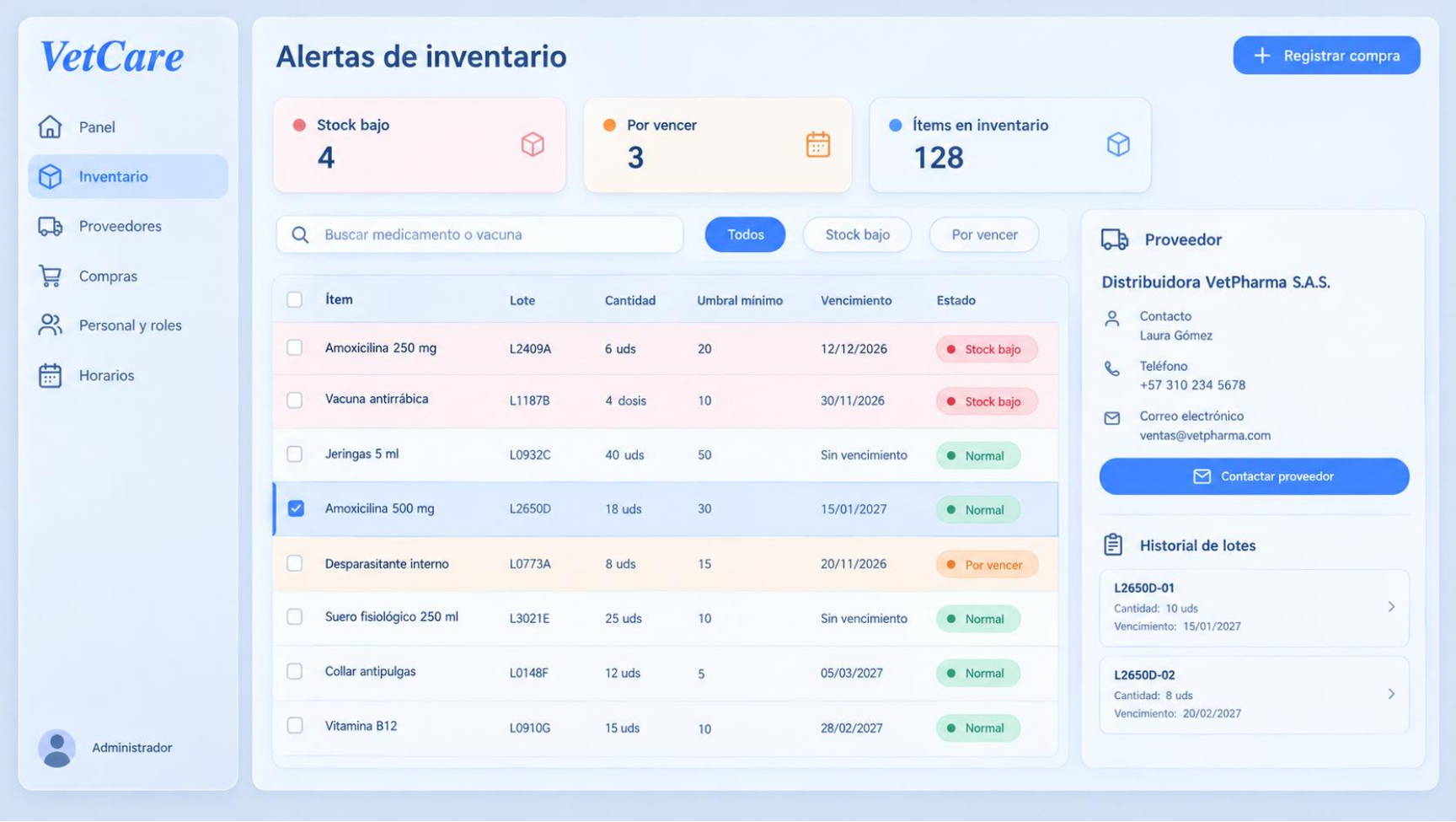
Visual Hospitalization Map and Rounds Log



Clinical Record and Medical Consultation



Appointment and Room Management



Pharmacy and Inventory Dashboard

6. References

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7. Use of Artificial Intelligence

In the UX/UI design phase of this project, Generative Artificial Intelligence tools (such as Stitch and Large Language Models) were utilized to assist in creating the platform's visual mockups and wireframes. The project team evaluated all AI-generated layouts, and refined the interface components to ensure full alignment with the functional requirements, Role-Based Access Control (RBAC) constraints, and user story acceptance criteria.